



Effect of PCM thickness on the performance of the finned PV/PCM system

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ABSTRACT

The electrical and thermal efficiency of photovoltaic (PV) panels depends mainly on the operating temperature, which is a major problem for this technology. For this reason, various studies have been developed to cool PV panels and consequently improve their efficiency. The use of phase change materials (PCMs) for cooling has attracted the attention of several researchers. This method consists of absorbing thermal energy in the form of latent heat by the PCM and thus decreasing the PV cell temperature. This paper investigates the temperature change of PCM, its liquid fraction, and the thermal behavior of a photovoltaic panel with and without a cooling system. This system is composed of an aluminum heat sink attached with fins and a layer of RT42 Paraffin-type PCM, placed on the backside of the photovoltaic panel. The present research focuses on the effect of PCM thickness on PV panel performance by comparing six different PCM thicknesses: 0.02 m, 0.03 m, 0.04 m, 0.05 m, 0.06 m, and 0.07 m, through a simulation using ANSYS fluent software. In order to make the study more realistic, a meteorological file of Oujda city (eastern Morocco) has been developed and added to the software using a UDF subroutine written in C++, the equations given have integrated the evolution of the ambient temperature and solar radiation as a function of time. The results show that integrating the PCM and fins to the PV panel decreases the PV cells temperature by 32 K compared to the conventional module. This temperature drop is due to the heat stored by the PCM, along with the use of fins that accelerated the heating and melting of PCM. Moreover, it was found that increasing its thickness further decreases the PV temperature. The thickness $e = 0.06$ m is selected as the most appropriate thickness for the best performance. Lower thicknesses melt quickly under the day sunlight and higher thicknesses do not reach full melting.

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1. Introduction

Currently, one of the most promising renewable energies is photovoltaic solar energy, as this technology uses clean and inexhaustible resources [1–3]. However, photovoltaic (PV) solar cells can convert only a part of the incident solar radiation, while the rest promotes its heating and consequently the increase of its temperature [4]. As a result, the efficiency of the PV panel, particularly, the PV cell is affected. Multiple cooling solutions are proposed in the literature to mitigate this negative effect of temperature. The methods used for PV cooling are generally classified into two categories: active and passive ones [5]. The main difference between them is that the latter does not require any mechanical device, while the first requires a pump or fan to circulate the coolant. Thus, although passive cooling methods have a lower cooling perfor-

mance than active methods [6], they have many advantages [7], including no operational cost [8], lower associated investment [9], and simplicity of operational aspects [10].

Due to the high amount of absorbed radiation converted to thermal energy, the electrical performance is negatively affected, as proven by the fact that a 274.15 K increase in the average temperature of the solar cell decreases its electrical conversion efficiency by about 0.4–0.5 % [11]. Therefore, it has become imperative to improve the performance of PV through cooling methods to recover a significant amount of wasted energy and regulate the operational temperature of photovoltaic systems. In the next two sections, we will give an overview of some passive and active cooling methods covered in the literature.

A Computational fluid dynamic (CFD) study of the impact of frame perforations on the passive cooling of PV modules was conducted by S. Nazari et al. [12]. The investigation considered several patterns of perforations introduced into the aluminum frame of the PV module, with the aim of comparing the resulting temperature

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distribution and power output with those of conventional frames. From the reviewed cases, an improvement of 0.62 % in electricity production was found by one of the proposed systems. **A. Bria et al. [13]** studied the temperature reduction of PV panels using a heat sink to increase the surface area and heat transfer rate. The proposed heat sink was proposed as an aluminum plate with fins attached to the back surface of the panel. A comparison was made between two types of fins: flat fins and lapping fins, as well as a study of the effect of lapping size on the cooling system. The results showed that the use of flat fins can reduce the average temperature of the panel by 18.16°K, while with fine lapping, the temperature reduction was decreased by 23.7°K. Thus, it was found that the temperature of the PV panel decreases as the lapping height increases. **S. Nizetic et al. [14]** conducted another research that reported A new and effective passive cooling strategy for the photovoltaic panel. The experiment involved several small PCM-filled containers and compared the performance of the system in each configuration with that of the conventional system. The results reveal that the proposed innovative PV-PCM passive cooling method can improve the performance by 10.7 %, while the conventional approach that uses a single full PCM container shows only a 2.5 % improvement. Another experimental study was conducted by **M. Dida et al. [15]** under some of the hottest weather conditions. The research team developed a new system capable of reducing the temperature by about 26 %, while ensuring uniformity of the operating temperature across the PV module, even at midday. Furthermore, in another passive cooling study, researchers **R. Rabie et al. [16]** extended the aluminum fins beyond the PCM container. The objective was to ensure that the temperature of the PV panel does not increase after the PCM has completely melted. The results reveal that the use of PCM regulated the temperature of the solar cells during the solid–liquid phase transition process. After melting, the capacity of PCM decreased, so the use of elongated aluminum fins extended the melting time, which prevented the temperature increase. **A. Bria et al. [17]** investigated the finned PV/PCM system for PV panel cooling as well as the effect of fins length on panel efficiency. The results show that the PV cell temperature decreases by 13.8°K with the cooling system and that increasing the fin length decrease the PV cell temperature. **T. Wongwuttanasatian et al. [18]** investigated different PCM container designs, including grooved, tube, and finned containers; the finned PCM box was found to be the most effective cooling method for the PV module. **A. Bria et al. [19]** studied the effect of different types of PCM namely RT58, RT42, and C22–C40 with a heat sink attached to the backside of the panel on their efficiency. The study showed that the integration of PCM decreases the temperature of the PV panel by 21.2°K, 26.1°K, and 18.3°K for RT58, RT42, and C22–C40 types, respectively.

To maintain the photovoltaic panels at low temperatures, various methods are available, with particular attention being paid to active methods. **H. Metwally et al. [20]** investigated an active cooling system to evaluate the annual performance of photovoltaic panels. They managed to keep the temperature of photovoltaic panels at a constant temperature for nearly 2 h and reduced the temperature in winter, spring, and summer to 295 K, 302 K, and 311 K, respectively. Another significant increase in solar cell efficiency was achieved with a new active cooling system proposed, fabricated, and studied by the team of **H.G. Teo et al. [21]**. The performance enhancement is in the range of 12–14 %, with the system essentially consisting of a parallel array of conduits with inlet and outlet collectors. Active cooling methods have also been applied in several research studies on the concentrator photovoltaic panel, such as the work done by **S. A. Zubeer et al. [22]** and **H. I. Elqady et al. [23]** in which they improved the electrical efficiency by reducing the panel temperature.

The main objective of this study is to investigate the conventional passive cooling of PV panels using PCM to enable efficient thermal management. The approach is to use a container attached to the back surface of the PV panel and fill it with PCM to regulate its operating temperature. The thickness of the PCM has been investigated, to determine the optimal amount of the material that should be used to regulate the operational temperature, such that all of the PCM used is beneficial and no more is needed. To perform this evaluation, the ANSYS Fluent software was used to numerically study the temperature distribution of the PV layers along with the PCM. Besides the melting fraction of the PCM.

2. Methodology

2.1. Physical model

ANSYS Fluent software based on the finite volume method was used to study temperature variation over time under the meteorological conditions of Oujda city (Eastern Morocco). The studied model is composed of a photovoltaic panel and a cooling system consisting of a PCM layer and an aluminum heat sink with flat fins placed on the backside of the panel (Fig. 1). The geometric dimensions and thermophysical properties of each layer of the system are presented in tables 1–2.

2.2. Mathematical model and boundary conditions

The present work aims to use PCM on the back surface of the PV panel, which makes it a simple cooling mechanism for efficient operation. The approach proposed in this paper is a passive PV-PCM cooling mechanism, which has no operational cost and can guarantee a uniform temperature on the PV panel. Since the conductivity of the PCM is low, fins were used to improve the heat exchange and increase the amount of thermal energy absorbed by the PCM. In addition, to ensure that the amount of PCM used is sufficient to absorb the thermal energy of the panel throughout the day, the thickness was studied. In the investigated system (PV/PCM), the heat transfer between the PV walls and the outside is mainly done by convection, except for the exterior of the PV exposed to the sun where both convection and radiation take place. The walls of the absorber and the PCM layer are considered adiabatic walls (Fig. 2).

The partial differential equations representing the governing equations of the system are as follows:

- Continuity equation:

$$\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} = 0 \quad (1)$$

- Equation of momentum:

(x) axis:

$$\frac{\partial}{\partial t}(U) + \frac{\partial}{\partial X}(U^2) + \frac{\partial}{\partial Y}(UV) = -\frac{\partial P}{\partial X} + P_r \left(\frac{\partial^2 U}{\partial X^2} + \frac{\partial^2 U}{\partial Y^2} \right) + R_a P_r \sin(\alpha) + S_x \quad (2)$$

(y) axis:

$$\frac{\partial}{\partial t}(V) + \frac{\partial}{\partial X}(UV) + \frac{\partial}{\partial Y}(V^2) = -\frac{\partial P}{\partial Y} + P_r \left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2} \right) - R_a P_r \cos(\alpha) + S_y \quad (3)$$

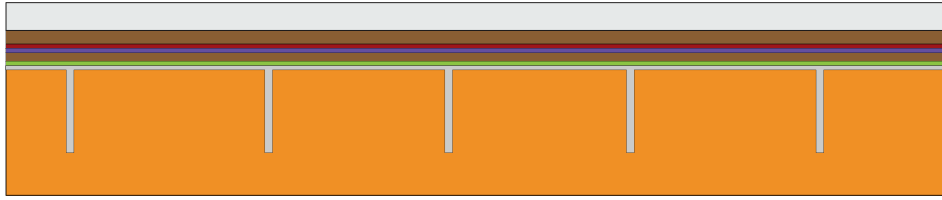


Fig. 1. The layers of the PV panel and the integrated cooling system [24–28].

Table 1

Geometric dimensions and thermophysical properties of different layers of the PV system.

Layers	Thickness(m)	Density (kg/m^3)	Thermal conductivity (W/mK)	Specific heat capacity (J/kgK)
Glass	0.004	2650	1.5	786
EVA	0.0005	960	0.35	2090
ARC	8.10^{-8}	4157	8.4	692
PV cells	0.0003	2330	100	700
Tedlar	0.0001	1200	0.2	1250
Absorber	0.003	2719	202.4	871

Table 2

Physical proprieties of Paraffin RT42 PCM.

PCM type	Paraffin RT42
Density (kg/m^3)	820
Thermal conductivity (W/mK)	0.2
Specific heat capacity (J/kgK)	2000
Melting heat (J/kg)	174,000
Melting temperature (K)	313.65

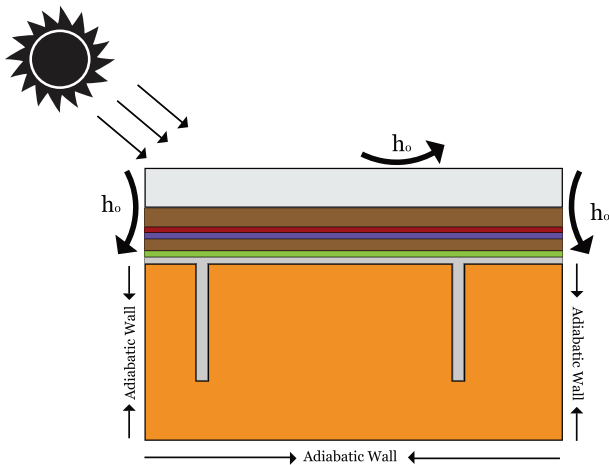


Fig. 2. Boundary conditions system.

• Equation of energy:

$$\frac{\partial}{\partial t}(\rho h) + \frac{\partial}{\partial x}(\rho u h) + \frac{\partial}{\partial y}(\rho v h) = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \frac{\partial T}{\partial y}\right) + S_h \quad (4)$$

Where:

U : the velocity in the X direction.

V : the velocity in the Y direction.

ρ : the density.

P : the pressure.

P_r : Prandtl number.

Re : Reynolds number.

k : the thermal conductivity.

S : the sources terms.

h : the specific enthalpy.

The sensible and total enthalpies h_s and H respectively, are presented in the following equation:

$$h_s = h_{ref} + \int_{T_{ref}}^T C_p dT \quad (5)$$

$$H = h_s + \Delta H \quad (6)$$

Where:

ΔH : the enthalpy changes due to the phase change ($\Delta H = \gamma L$).

h_{ref} : the reference enthalpy.

C_p : the specific heat.

L : the specific enthalpy of fusion (liquid state).

γ represents the liquid fraction during the phase change under the conditions presented in the equations.

$$\text{If } T < T_{solid} \gamma = \frac{\Delta H}{L} = 0 \quad (7)$$

$$\text{If } T_{solid} < T < T_{liquid} \gamma = \frac{\Delta H}{L} = \frac{T - T_s}{T_l - T_s} \quad (8)$$

$$\text{If } T > T_{liquid} \gamma = \frac{\Delta H}{L} = 1 \quad (9)$$

Concerning the convection losses in the walls of the photovoltaic panel, the heat coefficient is presented as:

$$h_0 = 5.67 + 3.86 V_{wind} \quad (10)$$

The pressure-based solver was used with the solidification and melting model to simulate the PCM and the second-order upwind scheme to solve the energy and momentum equations of the system. The pressure-velocity coupling is realized by the PISO algorithm and the PRESTO pressure scheme. For convergence, the under-relaxation factors used are 0.3, 0.7, and 0.9 for pressure, momentum, and energy, respectively. The convergence criteria have been fixed at 10^{-5} , 10^{-6} , and 10^{-8} for the continuity, momentum, and energy equation.

2.3. Climate data

To be more realistic, a climate data file of an August day was used. The data are extracted from the meteorological station located at the Mohamed 1er University of Oujda (Eastern Morocco) (Fig. 3), in order to integrate the evolution of the ambient temper-

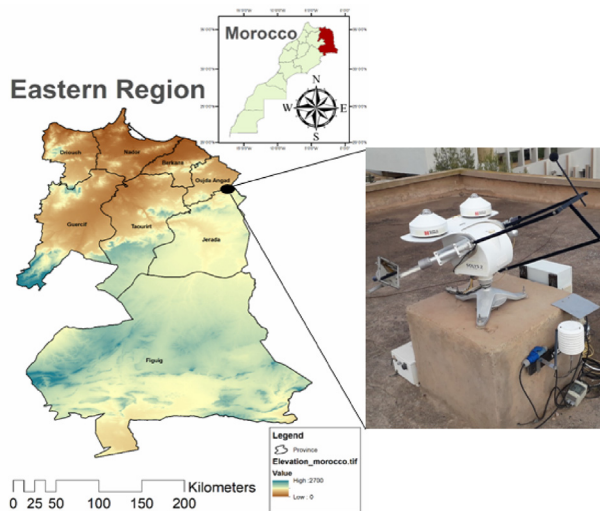


Fig. 3. Meteorological station installed at Mohammed First University Oujda (Eastern Morocco).

ature and solar radiation during the test day (Fig. 4). The station is equipped with pyranometers and pyrhemometers to measure irradiance data. The standard Kipp & Zonen CMP21 pyranometers have a spectral range of 285 to 2800 nm, a response time of 5 s, and a measurement error of 2 % [29]; while the pyrhemometer used has a spectral range of 200 to 4000 nm, the response time is less than 5 s and the measurement error is about 1 % [30]. For the temperature sensor, the operating temperature is between -223.15 and $+323.15$ K, with a high precision of ± 0.10 K. Moreover, for the integration of these parameters in ANSYS fluent, we have used UDF subroutines written in C++.

The ambient temperature and solar radiation equations are approximated using the mathematical model proposed by W. Chen and W. Liu [31] (Eq. (11)–(12)) and modified according to the climatic data of the test day (Eq. (13)–(14)).

$$T_{amb}(t) = T_{amb}^- + T_{ar} \cos\left(\frac{\pi}{12}(t - 14)\right) \quad (11)$$

$$G_{sun}(t) = \hat{G}_{sun} \sin\left(\frac{t-a}{b-a}\pi\right), \quad a < t < b \quad (12)$$

$$T_{amb}(t) = 25 + 6 \cos\left(\frac{\pi}{12}(t - 14)\right) \quad (13)$$

$$G_{sun}(t) = 965 \sin\left(\frac{t-6}{14}\pi\right), \quad 6 < t < 20 \quad (14)$$

2.4. Validation

To ensure the accuracy of our result, we performed two validations with experimental studies. One concerns an air PVT system [32], and the other concerns the melting of PCM (pure gallium) [34] that has been studied as well numerically by [33]. Fig. 5 represents the validation of the water outlet and PV cell temperatures. A good agreement between the experimental and simulation results is noticed, with a maximum error of 4.2 % for the solar cell temperature and 5 % for the outlet air temperature. Concerning the PCM, Fig. 6 shows the validation of the gallium melting front in the cavity. The maximum error between the experimental results and ours recorded for the melting front at 2 min, 6 min, 10 min, and 17 min is 7.8 %, 8.4 %, 15.7 %, and 2.2 %, respectively. While the maximum error between the experimental and numerical results of Mahmoudi et al. [33] at 2 min, 6 min, 10 min, and 17 min is 33.33 %, 16 %, 17.4 %, and 13.27 %, respectively.

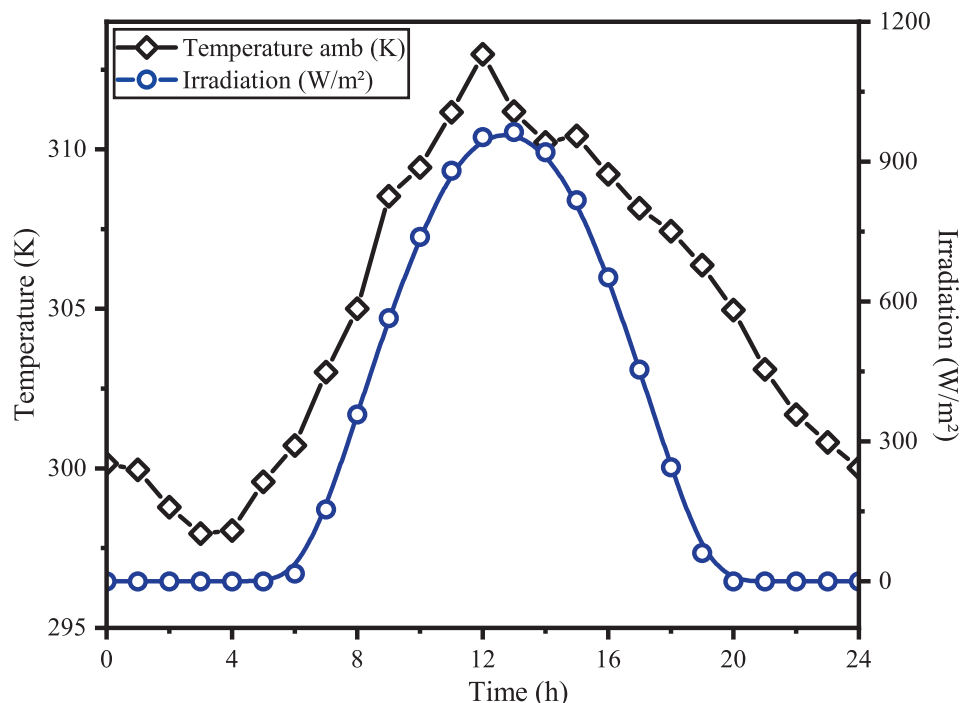


Fig. 4. The exported data graph.

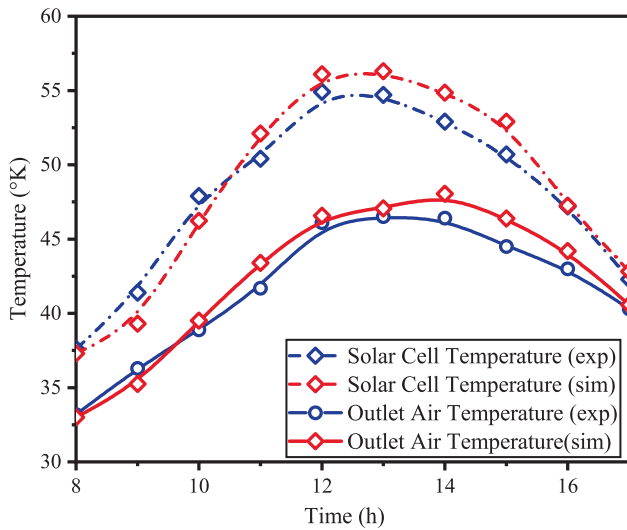


Fig. 5. Comparison of the PV cell temperature and the air outlet temperature with those of [32].

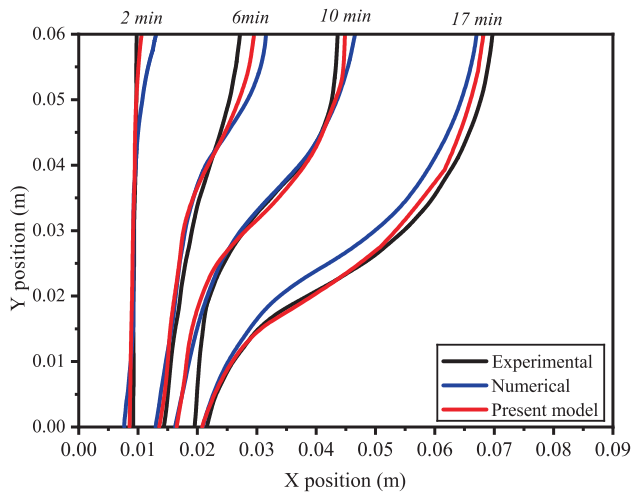


Fig. 6. Comparison of the melting front evolutions in the gallium melting cavity of numerical [33], experimental [34], and the present model.

2.5. Mesh independency test

The geometry representing the system was created in 2D using ANSYS Design Modeler. A quadratic mesh has been chosen for better convergence and to reduce the computing time. Fig. 7 shows the mesh grid of the system studied. A mesh independency study was performed to define the appropriate number of nodes required for this simulation. For this purpose, five mesh grids corresponding to a total number of nodes of 9014, 42554, 87243, 120456, 198052, and 232,541 were analyzed (Fig. 8). The average temperature of the PV cells was chosen as the parameter to be studied based on the mesh density. According to the figure, the convergence starts at a node number of 87243, which is considered sufficient for the con-

vergence as well as to decrease the computation time and the space in the memory.

3. Results and discussions

3.1. Effect of the cooling system on the temperature evolution of PV cells

Fig. 9 shows the temperature evolution of the PV cells with and without the cooling system during the test day with a PCM thickness of $e = 0.04$ m. A temperature increase is observed for both cases with and without the cooling system during the increase of solar radiation. The maximum temperature reached with and without cooling is 349 K at 12p.m. and 335 K at 6p.m., respectively. After this time the temperature decreases with the decrease of solar radiation. It is also noticed that before 4p.m. the PV cell's temperature with cooling system is lower than that without the cooling system, which is due to the thermal storage at the PCM level. On the contrary, after 4p.m., the PV cell's temperature of the system with cooling becomes higher than that of the system without cooling. This is caused by the liberation of the heat stored by the PCM.

3.2. Effect of PCM thickness on the PV panel cooling

This section investigates the effect of PCM thickness on PV panel cooling in order to achieve better performance. For this purpose, four thicknesses of PCM were used in this investigation: $e = 0.04$ m; 0.05 m; 0.06 m; 0.07 m. Fig. 10 shows the temperature evolution of the PV cells during the test day with the four different thicknesses. Before 1p.m., a negligible variation is noticed between the thicknesses of the PV cell temperature. However, after 1p.m., the temperature of the PV cell starts to vary, decreasing with increasing PCM thickness. The maximum PV cells temperature reached for the case of $e = 0.04$ m; 0.05 m; 0.06 m, and 0.07 m is respectively 335 K, 329.4 K, 327.8 K, and 327.4 K. This temperature drop is due to the proportional relationship between the thickness of the PCM and its total volume. If the volume of the PCM is larger, it takes longer to melt completely. Then, the PV panel temperature will increase slowly. This is confirmed by Fig. 11 which shows the evolution of the liquid fraction of the PCM during the test day. The liquid fraction increases by decreasing the thickness of the PCM which leads to the decrease of the PV cell's temperature. The PCM melting starts for the four PCM cases at 10 a.m. and reaches the total melting at 5p.m., 7p.m., and 9p.m. for the cases of $e = 0.04$ m, 0.05 m, and 0.06 m, respectively.

Fig. 12 shows the temperature and the liquid fraction contours of the finned PV/PCM system for the four different thicknesses ($e = 0.04$ m; 0.05 m; 0.06 m; 0.07 m). Concerning the liquid fraction, the blue and red colors indicate the solid ($fl = 0$) and liquid ($fl = 1$) states, respectively. The other different colors indicate the mushy zone ($0 < fl < 1$). It can be seen from the figure that the temperature decreases from the top surface of the PV panel to the PCM region for all cases of PCM thicknesses. We noticed as well that when we decrease the thickness of the PCM, it melts quickly due to the decrease in the amount of PCM used, resulting in a slower temperature increase. The Thickness $e = 0.06$ m represents the most appropriate thickness for the best performance with the required amount of PCM. Lower thicknesses melt quickly under the day's sun and higher thicknesses do not reach full melting.

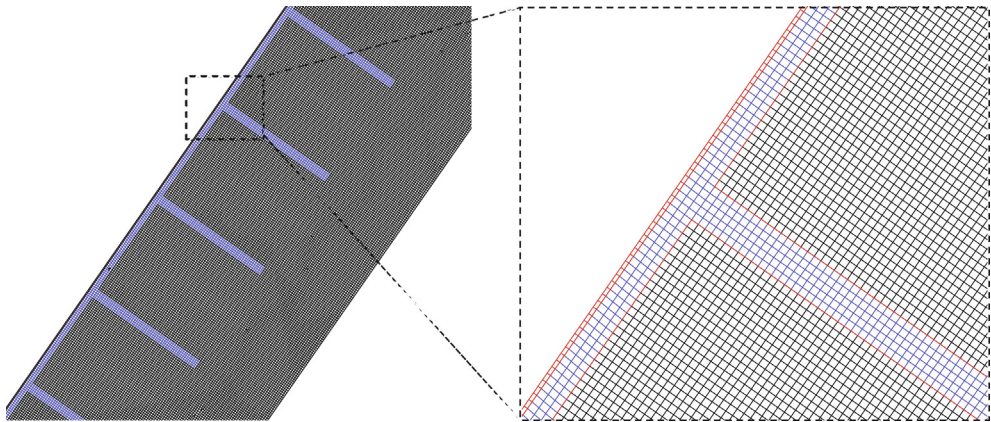


Fig. 7. Sections view of the mesh used in the finned PV/PCM system in the CFD simulation.

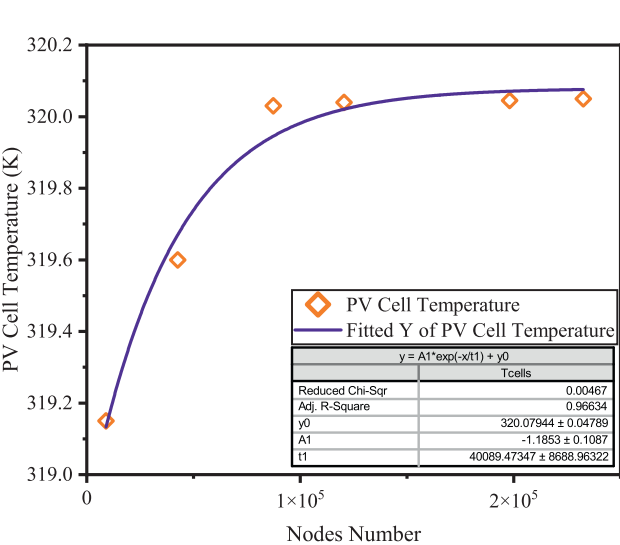


Fig. 8. Convergence test of mesh.

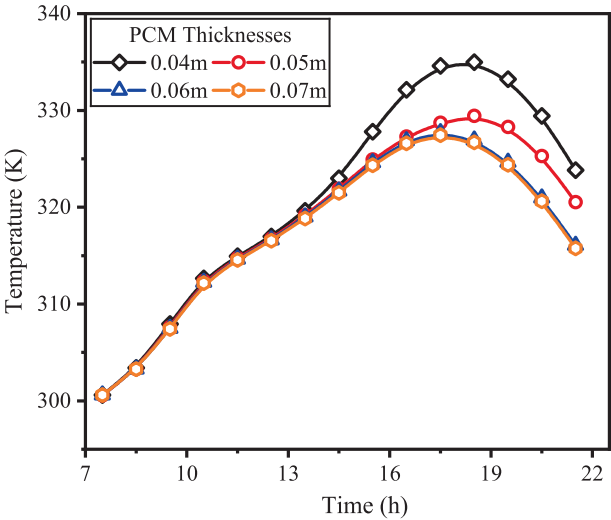


Fig. 10. Temperature evolution of PV cells under different PCM thicknesses during the test day.

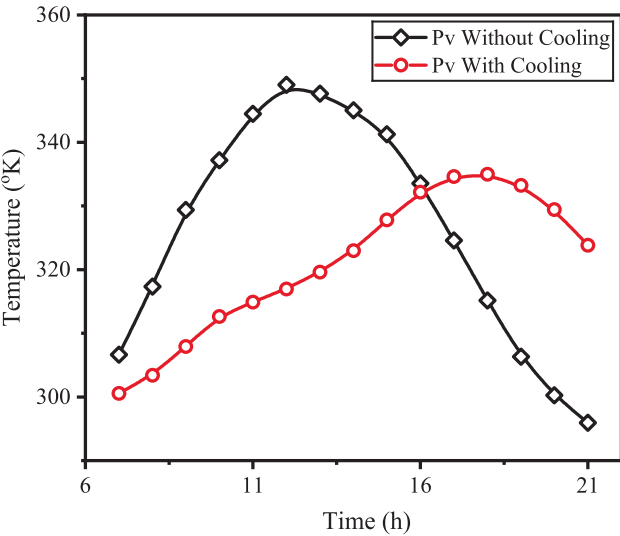


Fig. 9. The evolution of the PV cell temperature with and without cooling system during the test day.

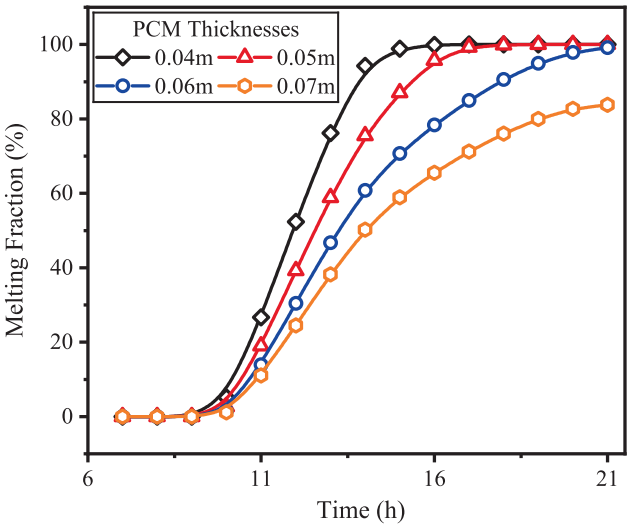


Fig. 11. Evolution of the melt fraction under different PCM thicknesses during the test day.

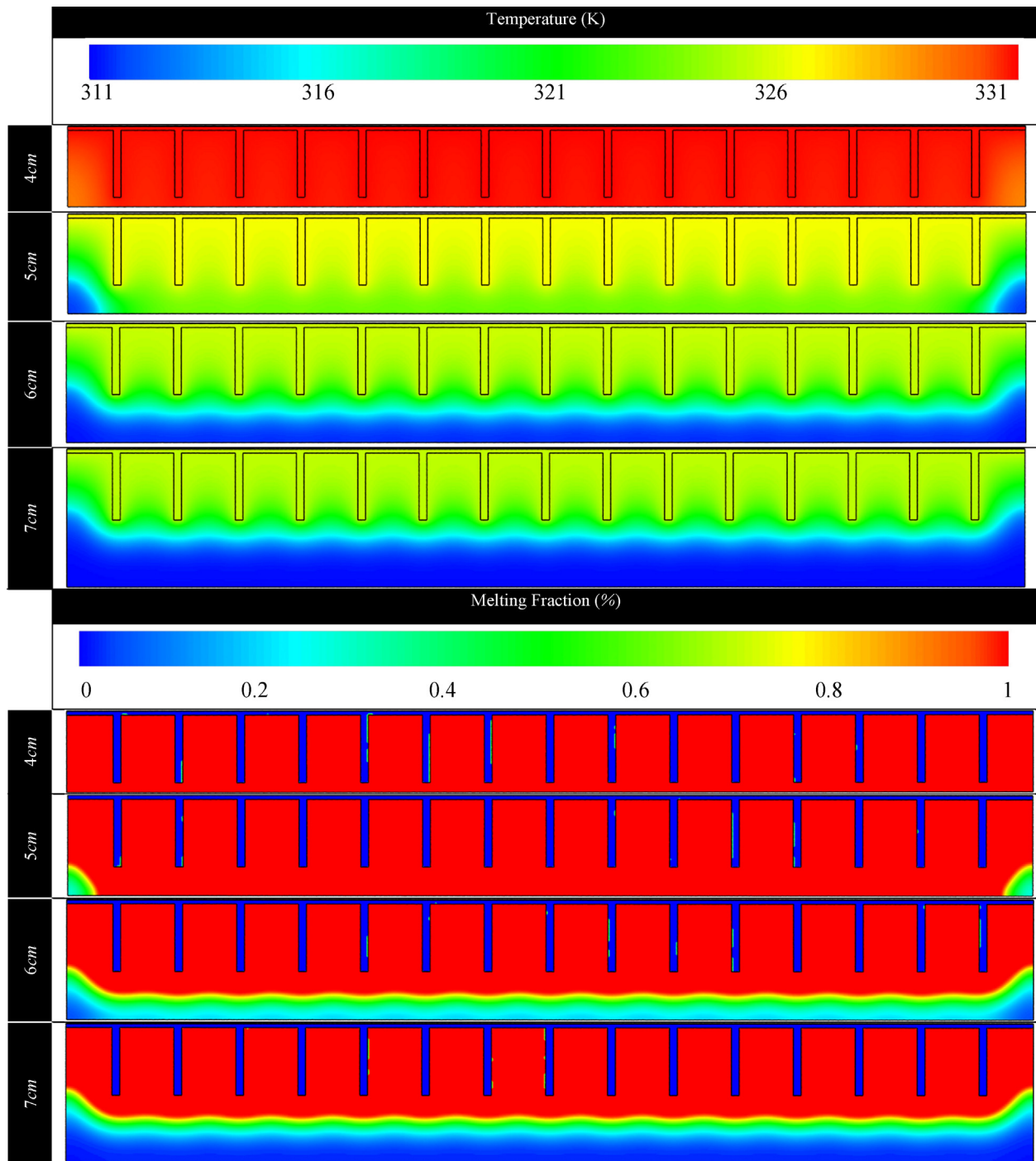


Fig. 12. Evolution of temperature and liquid fraction of the studied system under different thicknesses of the PCM during the test day.

3.3. Effect of cooling system on electrical parameters of PV panel

In this section, the effect of the cooling system and the PCM thickness on the electrical parameters of the panel were investigated.

Fig. 13 shows the maximum electrical output power for the cases without cooling system and with cooling system for the 0.04, 0.05, 0.06, and 0.07 m PCM thickness as a function of time. According to the figure, an increase in the maximum electrical power is observed for the cases with and without a cooling system

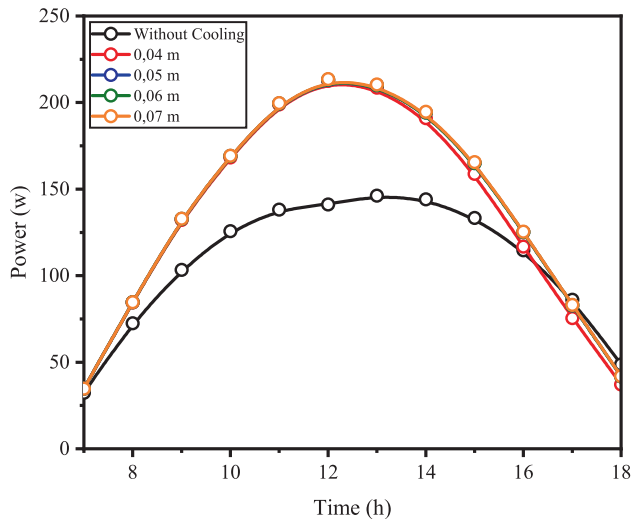


Fig. 13. The evolution of the maximum electrical power output with and without a cooling system during the test day.

during the increase of solar radiation. The maximum power reached during the test day without a cooling system is 146 W at 1 p.m and for the case with a cooling system for PCM thickness of 0.04, 0.05, 0.06, and 0.07 m is 212.48, 213, 213.34, and 213.47 W at 12 a.m, respectively. After this time, the maximum power decreases with the decrease of solar radiation. Therefore, we see an increase in the maximum power when the cooling system is integrated. So, It can be concluded that The thickness $e = 0.06$ m represents the most appropriate thickness for the best power with the necessary quantity of PCM.

Fig. 14 shows the effect of the cooling system on the P-V and I-V characteristics of a PV module at 12 a.m. The P-V characteristic represents the dependency between the electrical output power P of the PV panel and the output voltage V , while the I-V characteristic represents the dependency between the electrical current I and the output voltage V . The maximum electrical power for the case without cooling system is 141.17 W while for the case with cooling system is 213.34 W. So we can note that the increase in the maximum power of the PV panel with cooling was 72.2 W compared to the panel without cooling. With the cooling system, the short-circuit current of the PV cells changes slightly compared to the system without cooling, but the open-circuit voltage of the solar cells increases considerably, it can therefore be seen that the cooling system mainly influences the open-circuit voltage.

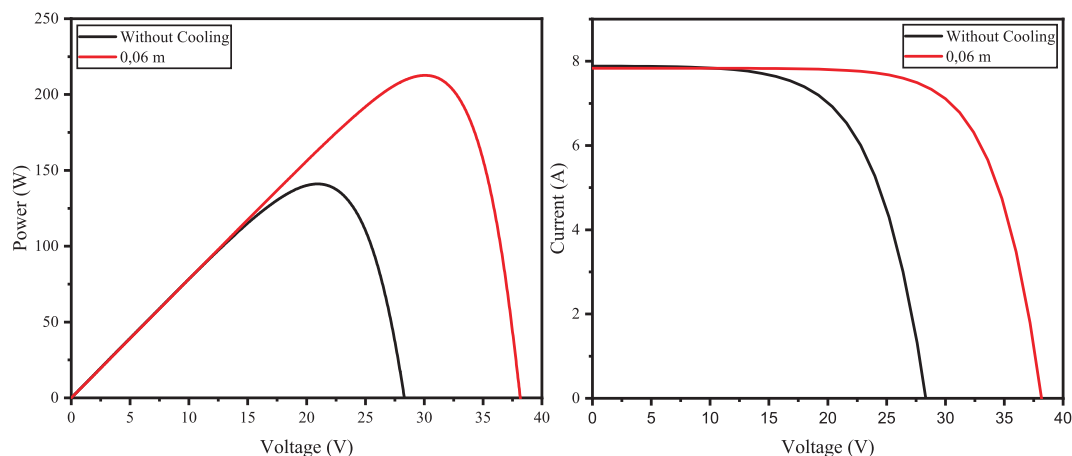


Fig. 14. Current-voltage and power-voltage characteristics of PV panel with and without cooling system at 12 a.m.

4. Conclusion

In this paper, the effect of phase change material with a heat sink on the cooling of photovoltaic panels was investigated under the climate conditions of Oujda city (Eastern Morocco). For this purpose, the Ansys Fluent software was used. The results show that the integration of the Paraffin (RT42) as a phase change material along with the fins in the PV panel reduces its temperature by up to 32 K compared to the conventional system. This is due to the heat stored by the PCM and the use of fins that accelerate the heating and melting of the PCM. Another investigation based on the effect of PCM thickness on PV cells temperature was performed. The results indicate that the decrease in PCM thickness decreases the PV cell's temperature. This temperature drop is due to the proportional relation between the PCM thickness and its total volume. The thickness $e = 0.06$ m represents the most appropriate thickness for the best performance with a maximum electric power output increase of 72.2 W. As recommendations, different points are proposed:

- Comparative study of finned PV/PCM systems, investigate different types of fins for enhanced PV performance.
- Investigation of the use of different layers with different types of PCM for better temperature uniformity of PV cells.
- Experimental study of the cooling of PV panels under different thicknesses of PCM.

CRediT authorship contribution statement

Abir Bria: Validation, Software, Investigation, Conceptualization. **Benyounes Raillani:** Conceptualization, Data curation, Formal analysis. **Dounia Chaatouf:** Software, Investigation, Formal analysis. **Mourad Salhi:** Methodology, Conceptualization, Resources. **Samir Amraoui:** Supervision, Project administration. **Ahmed Mezrhah:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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